LECTURE 1.4 · SHIP AND OCEAN SYSTEMS DESIGN

# Sizing from Regressions: Deadweight and Volume Carriers

Papanikolaou (2026) §1.3.6–1.3.7, §2.1–2.3, Appendix A · Session W1.3 · Lecture 50 min + Exercise 1.4 solved in class

**Luís Batista**

Instituto Superior Técnico / University of Lisbon

luis@veranavis.com

calendly.com/vnsd-luisbatista/meetme

# From the Owner’s Requirement to the First Dimensions

**1**

## Two routes

Deadweight carriers start from DWT; volume carriers from TEU or hold volume

**2**

## Deadweight route

The Appendix A fits on DWT, step by step, each value with its 95% band

**3**

## Volume route

From TEU (Kristensen) and from hold volume (the book’s example)

**4**

## Do they close?

Check the set: does weight equal buoyancy, are the ratios sound, is the ship inside the data? Then Exercise 1.4

**Read:** Papanikolaou (2026) §1.3.6–1.3.7, §2.1–2.3; Appendix A.1, A.3 and A.4

**Bring:** a laptop with Python or Excel, and Appendix A of the book

# Where You Start Depends on What Runs Out First

| Table 1.8 · Deadweight carriers | Basis for calculation |
|---|---|
| 0. Displacement Δ | From the given DWT (Table 2.1, App. A) |
| 1. Length L | Slenderness $L/\sqrt[3]{\nabla}$ (§2.3) |
| 2. Block coefficient CB | L and Fn = $V/\sqrt{(gL)}$ (§2.10) |
| 3. Beam B | Ratios L/B, B/T (§2.5–2.6) |
| 4. Draught T | Ratios B/T, L/T (§2.5, 2.8) |
| 5. Depth D | Required hold volume, ratio L/D (§2.7) |
| 6. CM, CP, CWP | From CB or Fn (§2.9–2.12) |

| Table 1.9 · Volume carriers | Basis for calculation |
|---|---|
| 1. Length L | Hold capacity (bale or net), or TEU number |
| 2. Block coefficient CB | L and Fn |
| 3. Beam B | L/B, or the same data as for L |
| 4. Depth D | B/D, or capacity/(L·B·D) |
| 5. Lightship WL | WL/(L·B·D) of similar ships, CB |
| 6. Deadweight DWT | Cargo, fuel and supplies |
| 7. Displacement Δ | WL + DWT |
| 8. Draught T | From Δ, L, B and CB |
| 9. CM, CP, CWP | From CB and Fn |

**Then both routes continue the same way · §1.3.7.1, steps 3–10**

**Power** → **Lines + GA** → **Weight balance** → **Hold volume** → **Freeboard** → **Stability** → **Cost** ↻ **about 3 loops**

# From Deadweight to Displacement

| Ship type | Size | DWT/Δ |
|---|---|---|
| Crude oil tankers | 25–500 kt | 0.78–0.88 |
| Bulk carriers | 20–200 kt | 0.74–0.87 |
| General cargo | 5–15 kt | 0.65–0.80 |
| **Container ships** | **10–165 kt** | **0.65–0.76** |
| Ro-Ro cargo | up to 16 kt | 0.50–0.60 |
| Ro-Pax ferries | L 85–120 m | 0.16–0.33 |
| Tugs | 0.5–3 MW | 0.20–0.40 |

$$\Delta = \frac{\mathrm{DWT}}{(\mathrm{DWT}/\Delta)}$$

$$\Delta = W_L + \mathrm{DWT}, \qquad W_L = \text{lightship}$$

Heavier cargo, larger ratio. Volume carriers have small ratios and a wide spread: start from Δ only with good data from sister ships.

For container ships Appendix A gives (Fig. A.29):

$$\mathrm{DWT}/\Delta = 0.5608\,\mathrm{DWT}^{0.024}$$

about 0.70–0.72.

**This table** places the ship in its family and gives a first Δ. **Next slide:** the Appendix A fits for one ship type give Δ and every main dimension, sized to the DWT, each with a 95% band.

# The Appendix A Fits for Deadweight Carriers

| Quantity | Figs. tank · bulk | Tankers (A.4) | Bulk carriers (A.1) |
|---|---|---|---|
| **DWT/Δ** | A.51 · A.1 | $\mathrm{DWT}/\Delta = 0.5412\,\mathrm{DWT}^{0.0390}$ | $\mathrm{DWT}/\Delta = 0.4668\,\mathrm{DWT}^{0.0530}$ |
| **Δ** | A.52 · A.2 | $\Delta = 4774 + 1.1213\,\mathrm{DWT}$ | $\Delta = 2.214\,\mathrm{DWT}^{0.9439}$ |
| **L (LBP)** | A.53 · A.3 | $L = 6.864\,\mathrm{DWT}^{0.3072}$ | $L = 7.603\,\mathrm{DWT}^{0.3002}$ |
| **B** | A.54 · A.4 | $B = 0.9624\,\mathrm{DWT}^{0.3249}$ | $B = 1.0559\,\mathrm{DWT}^{0.3097}$ |
| **D** | A.55 · A.5 | $D = 0.4591\,\mathrm{DWT}^{0.3306}$ | $D = 0.5843\,\mathrm{DWT}^{0.3108}$ |
| **T** | A.56 · A.6 | $T = 0.4501\,\mathrm{DWT}^{0.3031}$ | $T = 0.4807\,\mathrm{DWT}^{0.2983}$ |
| **CB** | — · A.10 | $C_B = 1.06 - 1.68\,F_n$ | $C_B = 0.5158\,\mathrm{DWT}^{0.0426}$ |
| **Lightship** | A.58 · A.9 | $W_L = 2.919\,\mathrm{DWT}^{0.7555}$ | $W_L/\Delta = 0.6564 - 0.04492\,\ln\mathrm{DWT}$ |

DWT in t, dimensions in m. Tankers have no CB fit: take it from the speed, Horn (Table 2.15), as the course does for the reference ship.

### Why we need them

At the start only DWT is known: these fits give a whole first ship in minutes, in line with the fleet.

### Then they bound and check

The 95% bands bound the optimization (D6) and check your ship model against real ships (D5).

### What they are not

A design method: each fit stands alone, so the set must still be made consistent (Lecture 1.5).

# A 60,000 t Bulk Carrier, Step by Step

A bulk carrier of 60,000 t DWT at 14 kn (an example, not one of our ships): put the DWT in each fit of slide 5.

| | | |
|---|---|---|
| 1 | Δ from the DWT/Δ ratio (Fig. A.1) | $\frac{\mathrm{DWT}}{\Delta} = 0.4668 \times 60{,}000^{0.0530} = 0.8363 \;\Rightarrow\; \Delta = \frac{60{,}000}{0.8363} = 71{,}740\ \mathrm{t}$ |
| 2 | Length (Fig. A.3) | $L = 7.603 \times 60{,}000^{0.3002} = 206.7\ \mathrm{m}$ |
| 3 | Beam and draught (Figs. A.4, A.6) | $B = 1.0559 \times 60{,}000^{0.3097} = 31.87\ \mathrm{m},\qquad T = 0.4807 \times 60{,}000^{0.2983} = 12.80\ \mathrm{m}$ |
| 4 | Depth and CB (Figs. A.5, A.10) | $D = 0.5843 \times 60{,}000^{0.3108} = 17.85\ \mathrm{m},\qquad C_B = 0.5158 \times 60{,}000^{0.0426} = 0.824$ |
| 5 | Lightship (Fig. A.9) | $\frac{W_L}{\Delta} = 0.6564 - 0.04492 \ln 60{,}000 = 0.162 \;\Rightarrow\; W_L = 0.162 \times 71{,}740 = 11{,}640\ \mathrm{t}$ |

Seven separate fits, one per quantity, give a complete first ship in minutes. Each comes from its own scatter of real ships, so each value has its own **band** (next slide), and together they **need not balance** (slide 10).

# Reading Appendix A: a Value and a Band

**Fig. A.3**

See Papanikolaou (2026), Appendix A

Fig. A.3 · Bulk carriers. Outer purple curves: 95% prediction band, about ±11% on L.

### The 60,000 t bulk carrier

| Item | Mean | 95% band |
|---|---|---|
| Δ | 71,740 t | ≈ 69,600–73,900 t |
| L | 206.7 m | ≈ 184–230 m |
| B | 31.9 m | ≈ 28.4–35.4 m |
| T | 12.8 m | ≈ 11.4–14.2 m |
| D | 17.9 m | ≈ 15.9–19.8 m |
| CB | 0.824 | ≈ 0.78–0.87 |
| Lightship | 11,640 t | ≈ 9,300–13,600 t |

**Note 1 · The band**

The mean is a start, not an answer: **95% of real ships lie inside the band**. A design outside it needs a reason.

**Note 2 · Displacement from the mean dimensions**

$\Delta = \rho C_B L B T = 1.025 \times 0.824 \times 206.7 \times 31.87 \times 12.80 = 71{,}220\ \mathrm{t}$

Against 71,740 t from the DWT fit: **−0.7 %**. Close, but not equal (slide 10).

# A 2,000 TEU Container Ship, Step by Step

**Fig. A.47**

See Papanikolaou (2026), Appendix A

**Fig. A.48**

See Papanikolaou (2026), Appendix A

Figs. A.47–A.48 (Kristensen): curves, no printed formula.

| | | |
|---|---|---|
| 1 | TEU → L (Fig. A.47) | 2,000 TEU → L ≈ 182 m |
| 2 | L → B and T (Fig. A.48) | B ≈ 30 m · T ≈ 10.2 m |
| 3 | TEU → payload → DWT (Fig. A.49) | $\mathrm{DWT} = \frac{\text{payload}}{0.75} = \frac{20{,}000}{0.75} = 26{,}700\ \mathrm{t}$ |
| 4 | L → service speed (Fig. A.50) | V ≈ 20.3 kn · Fn = 0.25 |
| 5 | Δ from DWT (Fig. A.29) | $\frac{\mathrm{DWT}}{\Delta} = 0.5608 \times 26{,}700^{0.024} = 0.716 \;\Rightarrow\; \Delta = 37{,}230\ \mathrm{t}$ |
| 6 | CB that results: check it | $C_B = \frac{\Delta}{\rho L B T} = \frac{37{,}230}{1.025 \times 182 \times 30 \times 10.2} = 0.65$ |

Here the order is reversed: **size from the boxes**, weights after. CB 0.65 against Horn’s 0.645 at Fn 0.25: the set is plausible.

# Sizing from Hold Volume

**Fig. 2.26**

See Papanikolaou (2026), Chapter 2

Fig. 2.26 · Follow the dashed line from 20,000 m³ (right) to L, B and D (top left).

**Given**

**Required hold volume:** ∇REQ = 20,000 m³ (grain +1–2 %, bale +11–12 %)

**Ratios of similar ships:** L/B = 6.5 · B/D = 1.8

**Engine room:** ¾ of the length aft

**Block coefficient at depth D:** CBD ≈ 0.70, from CB at the draught (inset, bottom left)

1. Read the hull volume below the deck: ∇H ≈ 27,560 m³, engine room ∇M ≈ 7,560 m³
2. Read the product at CBD 0.70: L·B·D ≈ 38,760 m³
3. **Read L, B, D: LPP = 139.8 m, B = 21.5 m, D = 12.0 m**
4. Check with the ratios:

$$B = \left[\frac{(L\,B\,D)\,(B/D)}{(L/B)}\right]^{1/3} = \left[\frac{38{,}760 \times 1.8}{6.5}\right]^{1/3} = 22.1\ \mathrm{m}$$

The formula gives B = 22.1, L = 143.4, D = 12.3 m: within 3 % of the diagram. A diagram is read to a few percent, like a regression band.

# Three Checks Before You Trust the Numbers

**1**

### Does the set close?

Each fit stands alone. Compare ρ·CB·L·B·T with Δ, and Δ − DWT with the lightship fit. Bulk carrier: −0.7% and −0.9%.

**2**

### Are the ratios sound?

L/B, B/T, L/D and Fn at the service speed against similar ships. Bulk carrier: 6.49, 2.49, 11.6; Fn = 0.16 at 14 kn.

**3**

### Is the ship inside the data?

A fit outside its data still gives a number, but not an answer. Outside? Change the source: another family, Kristensen’s TEU fits, your D2 ships.

**Regressions alone do not close the ship**

Each fit is made on its own, so buoyancy and weight need not agree: 0.7 % here, more when CB comes from another source. Exercise 1.4 measures the gap for the reference ship; Lecture 1.5 closes it with the design equation.

# Solved: the Reference Ship from the Tanker Fits

Oil/chemical tanker, DWT 37,600 t, 14.5 kn, draught at most 11.5 m (0.6, slide 2) · equations on slide 5 · ρ = 1.025 t/m³

**1** **Δ from DWT (Figs. A.51, A.52)**

$$\frac{\mathrm{DWT}}{\Delta} = 0.541\,\mathrm{DWT}^{\,0.039} = 0.817 \quad \Rightarrow \quad \Delta = \frac{37{,}600}{0.817} = 46{,}050\ \mathrm{t}$$

Fig. A.52: $\Delta = 4774 + 1.1213 \times 37{,}600 = 46{,}935\ \mathrm{t}\ \ (+1.9\,\%)$

**2** **L, B, T, D (Figs. A.53–A.56)**

$$L = 174.7,\ \ B = 29.50,\ \ T = 10.97,\ \ D = 14.95\ \mathrm{m} \ \Rightarrow\ \frac{L}{B} = 5.92,\ \ \frac{B}{T} = 2.69,\ \ \frac{B}{D} = 1.97$$

**3** **CB at 14.5 kn (Horn, Table 2.15)**

$$F_n = \frac{V}{\sqrt{g\,L}} = \frac{14.5 \times 0.5144}{\sqrt{9.81 \times 174.7}} = 0.180 \quad \Rightarrow \quad C_B = 1.06 - 1.68\,F_n = 0.757$$

**4** **Lightship (Fig. A.58)**

$$W_L = 2.919 \times 37{,}600^{0.7555} = 8{,}350\ \mathrm{t} \ \ \text{vs} \ \ \Delta - \mathrm{DWT} = 8{,}450\ \mathrm{t}\ \ (-1.2\,\%)$$

**5** **Does the set close?**

$$\rho\,C_B\,L\,B\,T = 1.025 \times 0.757 \times 174.7 \times 29.50 \times 10.97 = 43{,}880\ \mathrm{t}\ \ (-4.7\,\%)$$

**6** **Ratios, Fn, draught limit**

$$\frac{L}{D} = 11.7, \quad F_n = 0.180, \quad T = 10.97\ \mathrm{m} < 11.5\ \mathrm{m}$$

**Fig. A.51**
See Papanikolaou (2026), Appendix A

**Fig. A.53**
See Papanikolaou (2026), Appendix A

**Fig. A.54**
See Papanikolaou (2026), Appendix A

**Fig. A.55**
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**Fig. A.56**
See Papanikolaou (2026), Appendix A

**Fig. A.58**
See Papanikolaou (2026), Appendix A

Figs. A.51, A.53–A.56, A.58 · dashed: DWT 37,600 t

The fits give a plausible ship, but **buoyancy misses Δ by 4.7 %**. Lecture 1.5 closes the gap. **Your project:** repeat this for your ship (D2).

# The Story in Four Lines

**1 · Two routes.** Deadweight carriers start from DWT; volume carriers from TEU or hold volume, with T last.

**2 · Deadweight route.** Put the DWT in each Appendix A fit: a value and a 95% band, about ±11% on L, B, T, D.

**3 · Volume route.** Size from the boxes or the hold volume (Kristensen, Fig. 2.26); weights follow.

**4 · Checks.** Closure, ratios, data range. Regressions alone do not close the ship: Lecture 1.5.

**Next, same day (W1.4):** the design equation and Normand’s method; Exercise 1.5 on the reference ship. Read §2.13 and Appendix C.